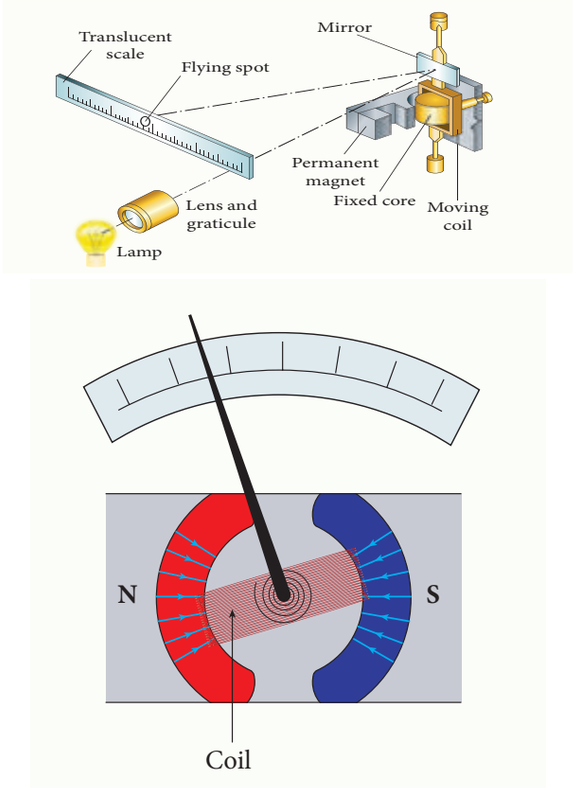


Figure 3.66 Moving coil galvanometer – its parts

phosphor bronze and the lower end of the coil is connected to a hair spring which is also made up of phosphor bronze. In a fine suspension strip W, a small plane mirror is attached in order to measure the deflection of the coil with the help of lamp and scale arrangement. The other end of the mirror is connected to a torsion head T. In order to pass electric current through the galvanometer, the suspension strip W and the spring S are connected to terminals.

Working

Consider a single turn of the rectangular coil PQRS whose length be l and breadth b. PQ = RS = l and QR = SP = b. Let I be the electric current flowing through the rectangular coil PQRS as shown in Figure 3.67. The horse-shoe magnet has hemi - spherical magnetic poles which produces a

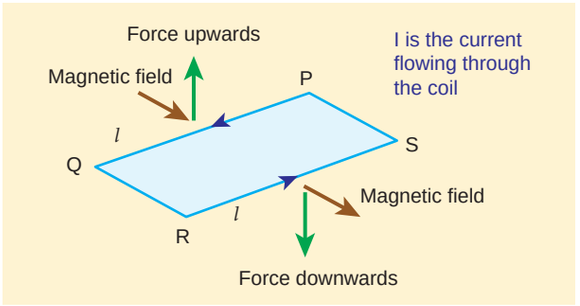


Figure 3.67 Force acting on current carrying coil

radial magnetic field. Due to this radial field, the sides QR and SP are always parallel to to the B-field (magnetic field) and experience no force. The sides PQ and RS are always parallel to the B-field and experience force and due to this, torque is produced.

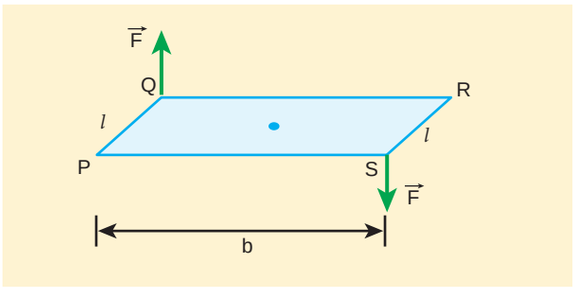


Figure 3.68 Deflection couple

For single turn, the deflection couple as shown in Figure 3.68 is

$$\tau = bF = bBIl = (lb)\,BI = ABI$$

since, area of the coil A = lb

For coil with N turns, we get

$$\tau = NABI \qquad (3.69)$$

Due to this deflecting torque, the coil gets twisted and restoring torque (also known as restoring couple) is developed. Hence the magnitude of restoring couple is proportional to the amount of twist θ (Refer Unit 10 of Std. XI Physics).Thus

$$\tau = K\theta \qquad (3.70)$$

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